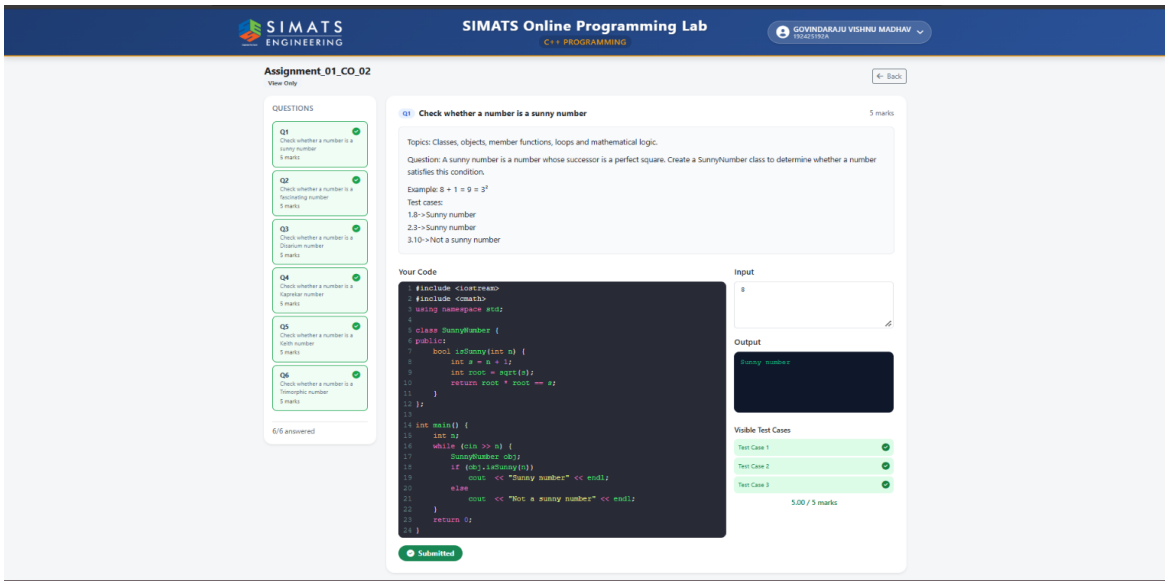
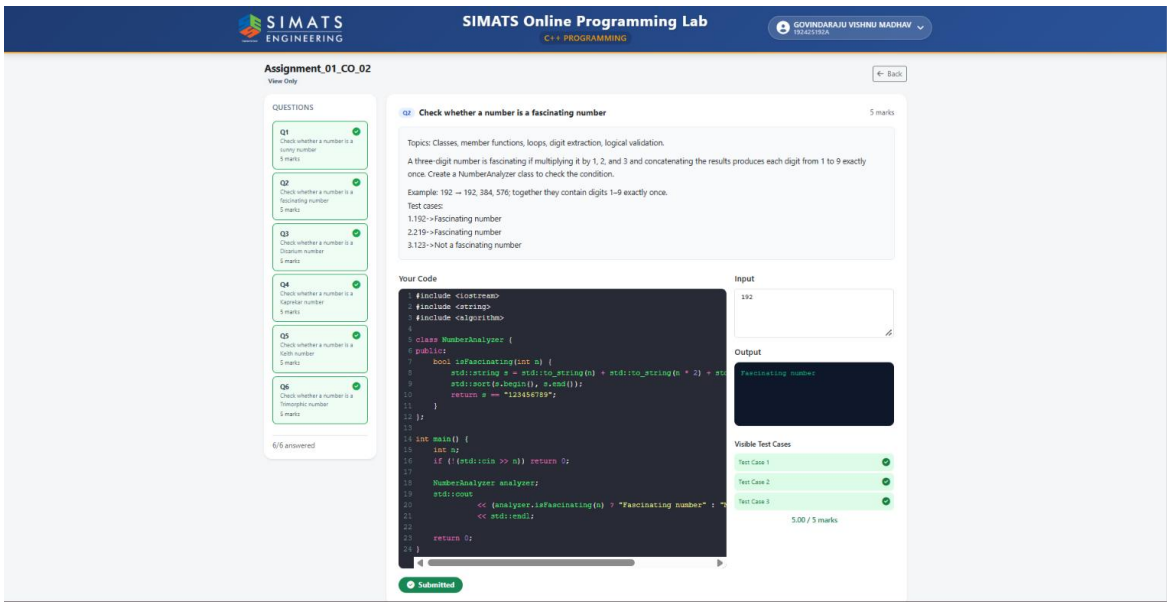
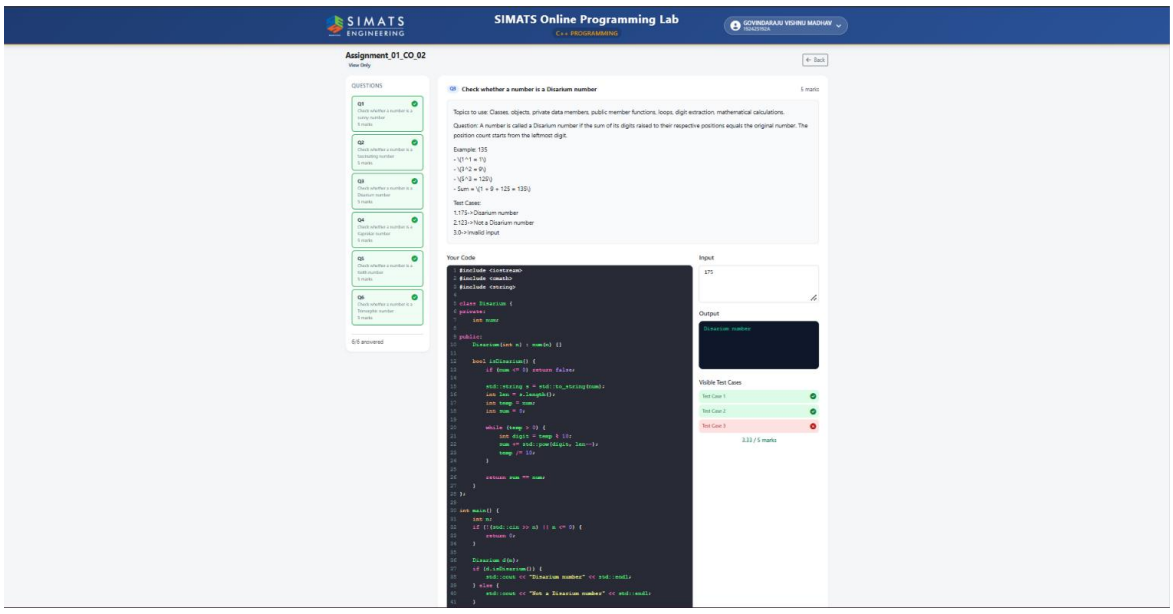
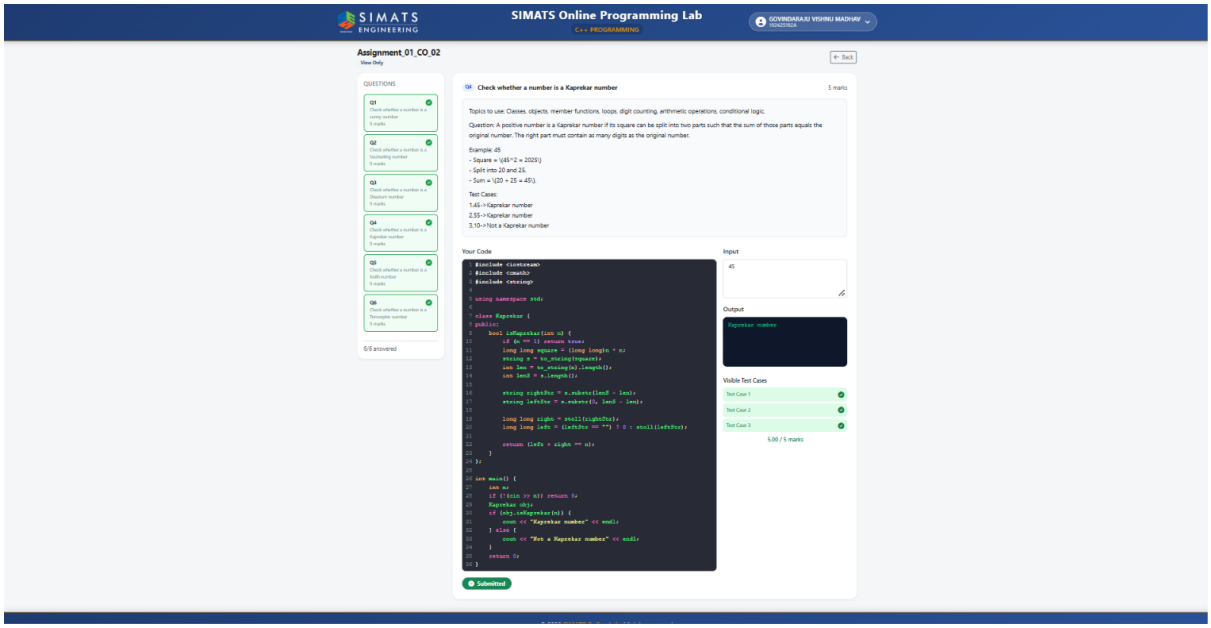
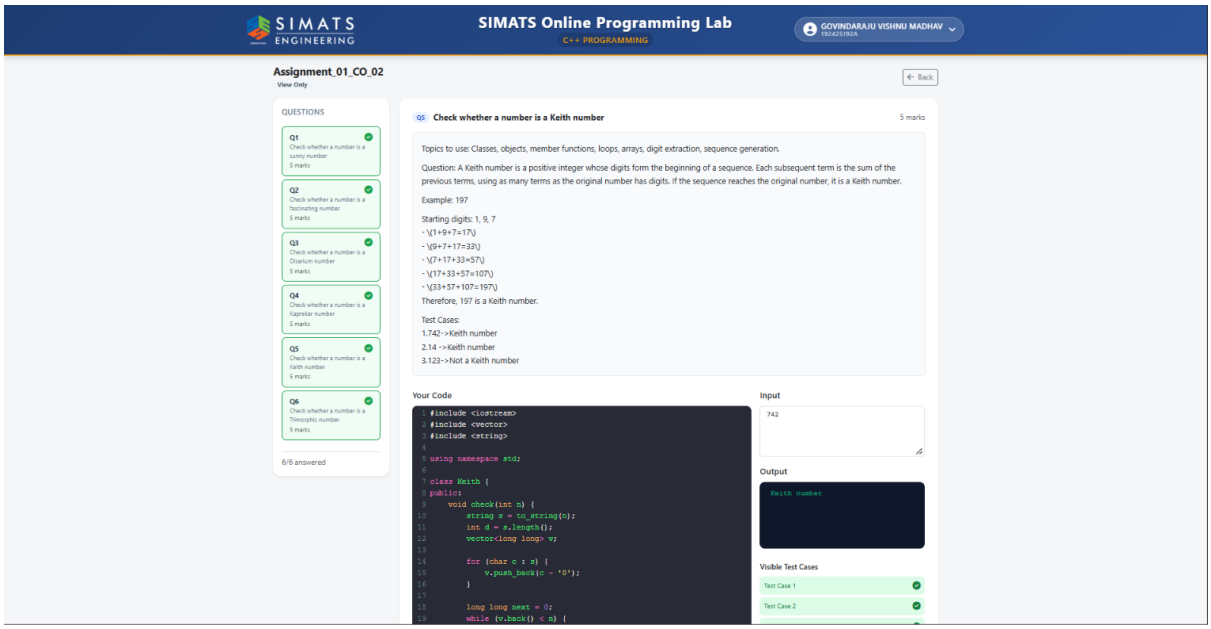
Co2 At 1

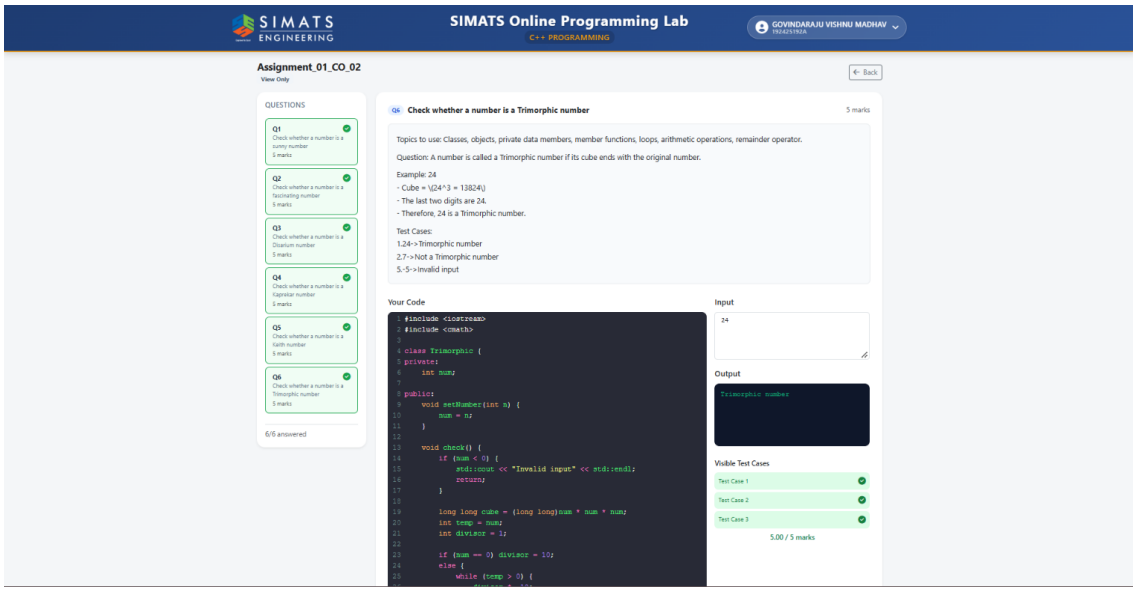

SIMATS ENGINEERING
SIMATS Online Programming Lab
C++ PROGRAMMING
GOVINDARAJU VISHNU MADHAV
Assignment_01_CO_02
View Only
Back
QUESTIONS
Q1 Check whether a number is a sunny number 5 marks
Topics: Classes, objects, member functions, loops and mathematical logic.
Question: A sunny number is a number whose successor is a perfect square. Create a SunnyNumber class to determine whether a number satisfies this condition.
Example: 8 + 1 = 9 = 3²
Test cases:
1.8->Sunny number
2.3->Sunny number
3.10->Not a sunny number
Your Code
Input
Output
Visible Test Cases
Test Case 1
Test Case 2
Test Case 3
5.00 / 5 marks
Submitted
6/6 answered


SIMATS ENGINEERING
SIMATS Online Programming Lab
C++ PROGRAMMING
GOVINDARAJU VISHNU MADHAV
Assignment_01_CO_02
View Only
Back
QUESTIONS
Q2 Check whether a number is a fascinating number 5 marks
Topics: Classes, member functions, loops, digit extraction, logical validation.
A three-digit number is fascinating if multiplying it by 1, 2, and 3 and concatenating the results produces each digit from 1 to 9 exactly once. Create a NumberAnalyzer class to check the condition.
Example: 192 → 192, 384, 576; together they contain digits 1–9 exactly once.
Test cases:
1.192->Fascinating number
2.219->Fascinating number
3.123->Not a fascinating number
Your Code
Input
Output
Visible Test Cases
Test Case 1
Test Case 2
Test Case 3
5.00 / 5 marks
Submitted
6/6 answered


SIMATS ENGINEERING
SIMATS Online Programming Lab
C++ PROGRAMMING
GOVINDARAJU VISHNU MADHAV
Assignment_01_CO_02
View Only
Back
QUESTIONS
Q5 Check whether a number is a Disarium number 5 marks
Topics to use: Classes, objects, private data members, public member functions, loops, digit extraction, mathematical calculations.
Question: A number is called a Disarium number if the sum of its digits raised to their respective positions equals the original number. The position count starts from the leftmost digit.
Example: 135
- 1^1 = 1
- 3^2 = 9
- 5^3 = 125
- Sum = 1 + 9 + 125 = 135
Test Cases:
1.175->Disarium number
2.123->Not a Disarium number
3.0->Invalid Input
Your Code
Input
Output
Visible Test Cases
Test Case 1
Test Case 2
Test Case 3
3.33 / 5 marks
6/6 answered


SIMATS ENGINEERING
SIMATS Online Programming Lab
C++ PROGRAMMING
GOVINDARAJU VISHNU MADHAV
Assignment_01_CO_02
View Only
Back
QUESTIONS
Q1
Q2
Q3
Q4
Q5
Q6
6/6 answered
Q4 Check whether a number is a Kaprekar number
5 marks
Topics to use: Classes, objects, member functions, loops, digit counting, arithmetic operations, conditional logic.
Question: A positive number is a Kaprekar number if its square can be split into two parts such that the sum of those parts equals the original number. The right part must contain as many digits as the original number.
Example: 45
- Square = \(45^2 = 2025\)
- Split into 20 and 25.
- Sum = \(20 + 25 = 45\).
Test Cases:
1.45->Kaprekar number
2.55->Kaprekar number
3.10->Not a Kaprekar number
Your Code
#include <iostream>
#include <cmath>
#include <string>
using namespace std;
class Kaprekar {
public:
bool isKaprekar(int n) {
if (n == 1) return true;
long long square = (long long)n * n;
string s = to_string(square);
int len = to_string(n).length();
int len2 = s.length();
string right = s.substr(len2 - len);
string left = s.substr(0, len2 - len);
long long r = stoll(right);
long long l = (left == "") ? 0 : stoll(left);
return (l + r == n);
}
};
int main() {
int n;
if (!(cin >> n)) return 0;
Kaprekar obj;
if (obj.isKaprekar(n)) {
cout << "Kaprekar number" << endl;
} else {
cout << "Not a Kaprekar number" << endl;
}
return 0;
}
Input
45
Output
Kaprekar number
Visible Test Cases
Test Case 1
Test Case 2
Test Case 3
5.00 / 5 marks
Submitted


SIMATS ENGINEERING
SIMATS Online Programming Lab
C++ PROGRAMMING
GOVINDARAJU VISHNU MADHAV
Assignment_01_CO_02
View Only
Back
QUESTIONS
Q1
Q2
Q3
Q4
Q5
Q6
6/6 answered
Q5 Check whether a number is a Keith number
5 marks
Topics to use: Classes, objects, member functions, loops, arrays, digit extraction, sequence generation.
Question: A Keith number is a positive integer whose digits form the beginning of a sequence. Each subsequent term is the sum of the previous terms, using as many terms as the original number has digits. If the sequence reaches the original number, it is a Keith number.
Example: 197
Starting digits: 1, 9, 7
- \(1+9+7=17\)
- \(9+7+17=33\)
- \(7+17+33=57\)
- \(17+33+57=107\)
- \(33+57+107=197\)
Therefore, 197 is a Keith number.
Test Cases:
1.742->Keith number
2.14 ->Keith number
3.123->Not a Keith number
Your Code
#include <iostream>
#include <vector>
#include <string>
using namespace std;
class Keith {
public:
void check(int n) {
string s = to_string(n);
int d = s.length();
vector<long long> v;
for (char c : s) {
v.push_back(c - '0');
}
long long next = 0;
while (v.back() < n) {
Input
742
Output
Keith number
Visible Test Cases
Test Case 1
Test Case 2


SIMATS ENGINEERING
SIMATS Online Programming Lab
C++ PROGRAMMING
GOVINDARAJU VISHNU MADHAV
Assignment_01_CO_02
View Only
Back
QUESTIONS
Q1
Q2
Q3
Q4
Q5
Q6
6/6 answered
Q6 Check whether a number is a Trimorphic number
5 marks
Topics to use: Classes, objects, private data members, member functions, loops, arithmetic operations, remainder operator.
Question: A number is called a Trimorphic number if its cube ends with the original number.
Example: 24
- Cube = \(24^3 = 13824\)
- The last two digits are 24.
- Therefore, 24 is a Trimorphic number.
Test Cases:
1.24->Trimorphic number
2.7->Not a Trimorphic number
3.-5->Invalid input
Your Code
#include <iostream>
#include <cmath>
class Trimorphic {
private:
int num;
public:
void setNumber(int n) {
num = n;
}
void check() {
if (num < 0) {
std::cout << "Invalid input" << std::endl;
return;
}
long long cube = (long long)num * num * num;
int temp = num;
int divisor = 1;
if (num == 0) divisor = 10;
else {
while (temp > 0) {
Input
24
Output
Trimorphic number
Visible Test Cases
Test Case 1
Test Case 2
Test Case 3
5.00 / 5 marks